

The oligodendrocyte-type-2 astrocyte cell lineage is specialized for myelination.

PMID: 3531873 | Indexed in PubMed

Astrocytes are one of the most numerous cell types in the vertebrate central nervous system (CNS) and yet their functions are largely unknown. In the rat optic nerve there are two distinct types of astrocyte: type-1 astrocytes develop from one type of precursor cell, and type-2 astrocytes develop from bipotential, oligodendrocyte-type-2 astrocyte (O-2A) progenitor cells, that initially give rise to oligodendrocytes (which make myelin in the CNS), and then to type-2 astrocytes. Type-1 astrocytes form the glial limiting membrane at the periphery of the optic nerve and are probably responsible for glial scar formation following nerve transection. The functions of type-2 astrocytes, which, like oligodendrocytes, are found mainly in tracts of myelinated axons throughout the CNS, are unknown. In this report we provide evidence that processes from type-2 astrocytes contribute to the structure of nodes of Ranvier, suggesting that the O-2A cell lineage is specialized for constructing myelin sheaths and nodes in the mammalian CNS.

Cellular Signal-Regulated Schwann Cell Myelination and Remyelination.

PMID: 31760634 | Indexed in PubMed

Increasing studies have demonstrated multiple signaling molecules responsible for oligodendrocytes and Schwann cells development such as migration, differentiation, myelination, and axo-glial interaction. However, complicated roles in these events are still poorly understood. This chapter focuses on well established intracellular signaling transduction and recent topics that control myelination and are elucidated from accumulating evidences. The underlying molecular mechanisms, which involved in membrane trafficking through small GTPase Arf6 and its activator cytohesins, demonstrate the crosstalk between well established intracellular signaling transduction and a new finding signaling pathway in glial cells links to physiological phenotype and essential role in peripheral nerve system (PNS). Since Arf family proteins affect the expression levels of myelin protein zero (MPZ) and Krox20, which is a transcription factor regulatory factor in early developmental stages of Schwann cells, Arf proteins likely to be key regulator for Schwann cells development. Herein, we discuss how intracellular signaling transductions in Schwann cells associate with myelination in CNS and PNS.

Regulators of Oligodendrocyte Differentiation.

PMID: 38503504 | Indexed in PubMed

Myelination has evolved as a mechanism to ensure fast and efficient propagation of nerve impulses along axons. Within the central nervous system (CNS), myelination is carried out by highly specialized glial cells, oligodendrocytes. The formation of myelin is a prolonged aspect of CNS development that occurs well into adulthood in humans, continuing throughout life in response to injury or as a component of neuroplasticity. The timing of myelination is tightly tied to the generation of oligodendrocytes through the differentiation of their committed progenitors, oligodendrocyte precursor cells (OPCs), which reside throughout the developing and adult CNS. In this article, we summarize our current understanding of some of the signals and pathways that regulate the differentiation of OPCs, and thus the myelination of CNS axons.

Oligodendrocytes: Myelination and Axonal Support.

PMID: 26101081 | Indexed in PubMed

Myelinated nerve fibers have evolved to enable fast and efficient transduction of electrical signals in the nervous system. To act as an electric insulator, the myelin sheath is formed as a multilamellar membrane structure by the spiral wrapping and subsequent compaction of the oligodendroglial plasma membrane around central nervous system (CNS) axons. Current evidence indicates that the myelin sheath is more than an inert insulating membrane structure. Oligodendrocytes are metabolically active and functionally connected to the subjacent axon via cytoplasmic-rich myelinic channels for movement of macromolecules to and from the internodal periaxonal space under the myelin sheath. This review summarizes our current understanding of how myelin is generated and also the role of oligodendrocytes in supporting the long-term integrity of myelinated axons.

Oligodendrocytes in Development, Myelin Generation and Beyond.

PMID: 31726662 | Indexed in PubMed

Oligodendrocytes are the myelinating cells of the central nervous system (CNS) that are generated from oligodendrocyte progenitor cells (OPC). OPC are distributed throughout the CNS and represent a pool of migratory and proliferative adult progenitor cells that can differentiate into oligodendrocytes. The central function of oligodendrocytes is to generate myelin, which is an extended membrane from the cell that wraps tightly around axons. Due to this energy consuming process and the associated high metabolic turnover oligodendrocytes are vulnerable to cytotoxic and excitotoxic factors. Oligodendrocyte pathology is therefore evident in a range of disorders including multiple sclerosis, schizophrenia and Alzheimer's disease. Deceased oligodendrocytes can be replenished from the adult OPC pool and lost myelin can be regenerated during remyelination, which can prevent axonal degeneration and can restore function. Cell population studies have recently identified novel immunomodulatory functions of oligodendrocytes, the implications of which, e.g., for diseases with primary oligodendrocyte pathology, are not yet clear. Here, we review the journey of oligodendrocytes from the embryonic stage to their role in homeostasis and their fate in disease. We will also discuss the most common models used to study oligodendrocytes and describe newly discovered functions of oligodendrocytes.

The roles of microglia and astrocytes in myelin phagocytosis in the central nervous system.

PMID: 36324281 | Indexed in PubMed

Myelination is an important process in the central nervous system (CNS). Oligodendrocytes (OLs) extend multiple layers to densely sheath on axons, composing the myelin to achieve efficient electrical signal conduction. The myelination during developmental stage maintains a balanced state. However, numerous CNS diseases including neurodegenerative and cerebrovascular diseases cause demyelination and disrupt the homeostasis, resulting in inflammation and white matter deficits. Effective clearance of myelin debris is needed in the region of demyelination, which is a key step for remyelination and tissue

regeneration. Microglia and astrocytes are the major resident phagocytic cells in the brain, which may play different or collaborative roles in myelination. Microglia and astrocytes participate in developmental myelination through engulfing excessive unneeded myelin. They are also involved in the clearance of degenerated myelin debris for accelerating remyelination, or engulfing healthy myelin sheath for inhibiting remyelination. This review focuses on the roles of microglia and astrocytes in phagocytosing myelin in the developmental brain and diseased brain. In addition, the interaction between microglia and astrocytes to mediate myelin engulfment is also summarized.

Myelination-independent functions of oligodendrocyte precursor cells in health and disease.

PMID: 37653126 | Indexed in PubMed

Oligodendrocyte precursor cells (OPCs) are a population of tissue-resident glial cells found throughout the CNS, constituting approximately 5% of all CNS cells and persisting from development to adulthood and aging. The canonical role of OPCs is to give rise to myelinating oligodendrocytes. However, additional functions of OPCs beyond this traditional role as precursors have been suggested for a long time. In this Perspective, we provide an overview of the multiple myelination-independent functions that have been described for OPCs in the context of neuron development, angiogenesis, inflammatory response, axon regeneration and their recently discovered roles in neural circuit remodeling.

Translating Molecular Approaches to Oligodendrocyte-Mediated Neurological Circuit Modulation.

PMID: 39061389 | Indexed in PubMed

The central nervous system (CNS) exhibits remarkable adaptability throughout life, enabled by intricate interactions between neurons and glial cells, in particular, oligodendrocytes (OLs) and oligodendrocyte precursor cells (OPCs). This adaptability is pivotal for learning and memory, with OLs and OPCs playing a crucial role in neural circuit development, synaptic modulation, and myelination dynamics. Myelination by OLs not only supports axonal conduction but also undergoes adaptive modifications in response to neuronal activity, which is vital for cognitive processing and memory functions. This review discusses how these cellular interactions and myelin dynamics are implicated in various neurocircuit diseases and disorders such as epilepsy, gliomas, and psychiatric conditions, focusing on how maladaptive changes contribute to disease pathology and influence clinical outcomes. It also covers the potential for new diagnostics and therapeutic approaches, including pharmacological strategies and emerging biomarkers in oligodendrocyte functions and myelination processes. The evidence supports a fundamental role for myelin plasticity and oligodendrocyte functionality in synchronizing neural activity and high-level cognitive functions, offering promising avenues for targeted interventions in CNS disorders.

The deletion of *dicer* in mature myelinating glial cells causes progressive axonal degeneration but not overt demyelination in adult mice.

PMID: 29726608 | Indexed in PubMed

Myelinating glial cells (MGCs), oligodendrocytes (OLs) in the central nervous system (CNS) and Schwann cells (SCs) in the peripheral nervous system (PNS), generate myelin sheaths that insulate axons. After myelination is completed in adulthood, MGC functions independent from myelin are required to support axon survival, but the underlying mechanisms are still unclear. Dicer is a key enzyme that is responsible for generating functional micro-RNAs (miRNAs). Despite the importance of Dicer in initiating myelination, the role of Dicer in mature MGCs is still unclear. Here, Dicer was specifically deleted in mature MGCs in 2-month old mice (PLP-CreERT; Dicer fl/fl) by tamoxifen administration. Progressive motor dysfunction was observed in the Dicer conditional knockout mice, which displayed hind limb ataxia at 3 months post recombination that deteriorated into paralysis within 5 months. Massive axonal degeneration/atrophy in peripheral nerves was responsible for this phenomenon, but overt demyelination was not observed in either the CNS or PNS. In contrast to the PNS, signs of axonal degeneration were not observed in the CNS of these animals. We induced a Dicer deletion in oligodendroglia at postnatal day 5 in NG2-CreERT; Dicer fl/fl mice to evaluate whether Dicer expression in OLs is essential for axonal survival. Dicer deletion in oligodendroglia did not cause motor dysfunction at the age of 7 months. Neither axonal atrophy nor demyelination was observed in the CNS. Based on our results, Dicer expression in SCs is required to maintain axon integrity in adult PNS, and Dicer is dispensable for maintaining myelin sheaths in MGCs.

The Role of Vesicle Trafficking and Release in Oligodendrocyte Biology.

PMID: 31782103 | Indexed in PubMed

Oligodendrocytes are a subtype of glial cells found within the central nervous system (CNS), responsible for the formation and maintenance of specialized myelin membranes which wrap neuronal axons. The development of myelin requires tight coordination for the cell to deliver lipid and protein building blocks to specific myelin segments at the right time. Both internal and external cues control myelination, thus the reception of these signals also requires precise regulation. In late years, a growing body of evidence indicates that oligodendrocytes, like many other cell types, may use extracellular vesicles (EVs) as a medium for transferring information. The field of EV research has expanded rapidly over the past decade, with new contributions that suggest EVs might have direct involvement in communications with neurons and other glial cells to fine tune oligodendroglial function. This functional role of EVs might also be maladaptive, as it has likewise been implicated in the spreading of toxic molecules within the brain during disease. In this review we will discuss the field's current understanding of extracellular vesicle biology within oligodendrocytes, and their contribution to physiologic and pathologic conditions.
